BASELINE VS TUNED MODEL EVALUATION

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WEEK :-3

DAY:-21

DATE:- 30.10.2026

BASELINE VS TUNED RESULTS

- Compared baseline models with tuned models to evaluate performance improvement.
- SOC estimation uses an XGBoost Regressor.
- Remaining charging time prediction uses a Random Forest Regressor.
- Model performance is evaluated using R^2 score and prediction error.

Day	Model	Target	Validation_MAE	Validation_RMSE	Validation_R2
Day 15	Linear Regression	SOC	15.70749655	21.03916515	0.3177928235
Day 15	Linear Regression	Remaining Charging	3274.076803	3667.171231	-5.45944368
Day 16	Random Forest Tune	SOC	9.072034605	11.6134019	0.7921365218
Day 16	Random Forest Tune	Remaining Charging	66.00901802	85.34167709	0.9965017158
Day 16	XGBoost Tuned	SOC	7.498428364	9.944823836	0.8475759604
Day 16	XGBoost Tuned	Remaining Charging	210.9542482	316.4501309	0.951900197
Day 17	Small ANN / MLP	SOC	11.61476189	17.77708665	0.5129422515
Day 17	Small ANN / MLP	Remaining Charging	85.53995298	117.117022	0.9934117089
Day 19	Random Forest - RF_	Remaining Charging	66.00901802	85.34167709	0.9965017158
Day 19	XGBoost - XGB_2_M	SOC	5.767199062	8.348844436	0.8925733541

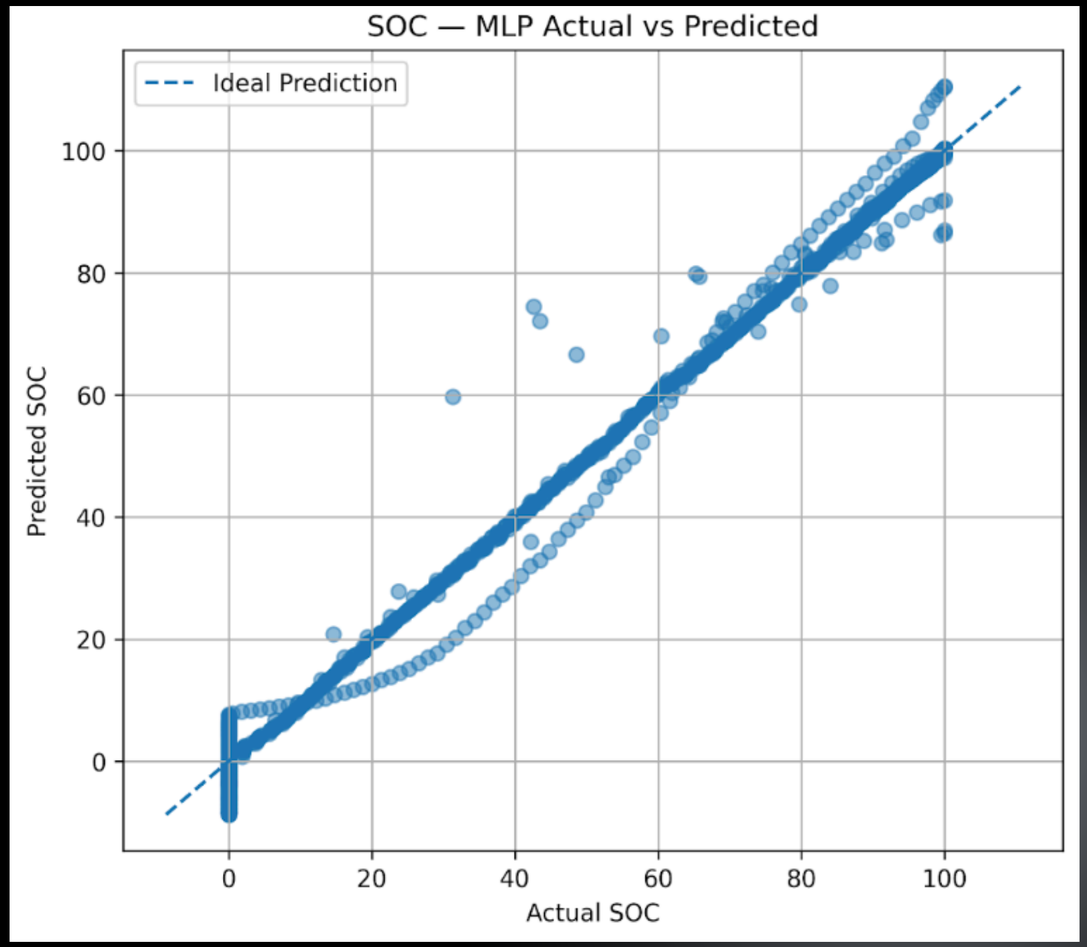
REPRESENTATIVE SUCCESSES AND FAILURES

- **Successful Predictions**

1. Predictions close to actual values indicate good model performance.
2. The models can learn patterns from battery voltage, current, temperature and other input features.

- **Prediction Errors**

1. Some predictions may differ from actual values.
2. Errors can vary with charging conditions and temperature.
3. Actual-versus-predicted values can help identify these case



REMAINING WEAKNESSES

- Prediction errors may still occur under certain charging conditions.
- Model performance can vary across different battery temperatures and operating conditions.
- Further improvement may be possible by analysing cases where predictions differ from actual values.

CONCLUSION

- Baseline and tuned models are compared to assess performance.
- Representative successes and failures are examined.
- Remaining weaknesses are identified for future improvement.

THANK YOU !